Student_dropout_prediction_project Public

About

This project addresses this challenge by leveraging machine learning techniques to analyze student data and predict dropout risk, with the aim of supporting data-driven decision-making and improving student retention strategies.

Readme

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Languages

- Jupyter Notebook 99.7%
- Python 0.3%

Suggested workflows

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Python Package using Anaconda

Configure

Create and test a Python package on multiple Python versions using Anaconda for package management.

Python package

Configure

Create and test a Python package on multiple Python versions.

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geoffreykemboi Merge pull request #29 from geoffreykemboi/master 519586f · yesterday 73 Commits

Student_dropout_prediction_project	adding index and dataset	last month
.DS_Store	readem ediited	5 days ago
.gitignore	Initial commit	last month
README.md	readme v1 editted	5 days ago
app.py	Add App.py file and deployment	yesterday
best_xgb.pkl	Add App.py file and deployment	yesterday
cleaned.csv	Add App.py file and deployment	yesterday
cleaned_notebook.ipynb	Add App.py file and deployment	yesterday
dropout_age_analysis.png	Innitial plots	2 weeks ago
gender_dropout_analysis.png	Innitial plots	2 weeks ago
index.ipynb	F2 score test	3 days ago
latest_notebook.ipynb	Modelling	4 days ago
model.pkl	Add App.py file and deployment	yesterday
student.csv	EDA Continuation	2 weeks ago
student_dropout_notebook_updated...	F2 score test	3 days ago

README

Analysis of Student Dropout Determinants in Kenyan Higher Education

Project Overview

This project addresses the critical challenge of student attrition within the Kenyan higher education system. By analyzing a dataset of over 100,000 students, the study identifies the complex interplay between academic performance, financial stability, and demographic factors.

Problem Identification

The core objective is to mitigate the socio-economic costs of student dropouts. The analysis focuses on:

- **Financial Stress:** Assessing the impact of tuition fee status and student loans.
- **Academic Preparation:** Comparing entry grades (High School) with university performance (GPA).
- **Engagement:** Evaluating how attendance and extracurricular involvement influence retention.

Tech Stack

- **Language:** Python
- **Key Libraries:** BalancedRandomForestClassifier, ColumnTransformer, ConfusionMatrixDisplay, LinearRegression, LogisticRegression, Pipeline, RFE, RandomForestClassifier, SMOTE, SimpleImputer, StandardScaler, XGBClassifier, accuracy_score,auc,roc_curve, chi2_contingency, datetime, matplotlib, numpy, pandas, seaborn, train_test_split, ttest_ind, warnings

- **Model:** XGBoost (Optimized via GridSearchCV)

Methodology

1. **Data Cleaning:** Imputation and handling of categorical variables.
2. **Feature Engineering:** Selection of key determinants for student success.
3. **Modeling:** Implementing an XGBoost classifier to predict dropout risk.
4. **Evaluation:** Assessing performance using ROC-AUC and Accuracy.

Key Results

The tuned XGBoost model produced the following results:

```
TUNED XGBOOST
Accuracy: 0.6974763406940063
ROC-AUC: 0.49535119493033175
```

Note: The model highlights that financial status (being a debtor) and academic performance are leading indicators of dropout risk.

Installation

1. Clone the repository:

   ```
   git clone https://github.com/your-username/student-dropout-analysis.git
   ```

2. Install required packages:

   ```
   pip install pandas numpy matplotlib seaborn scikit-learn xgboost
   ```

Usage

Open the `index.ipynb` file in Jupyter Notebook or Google Colab to view the full analysis, visualizations, and model training steps.